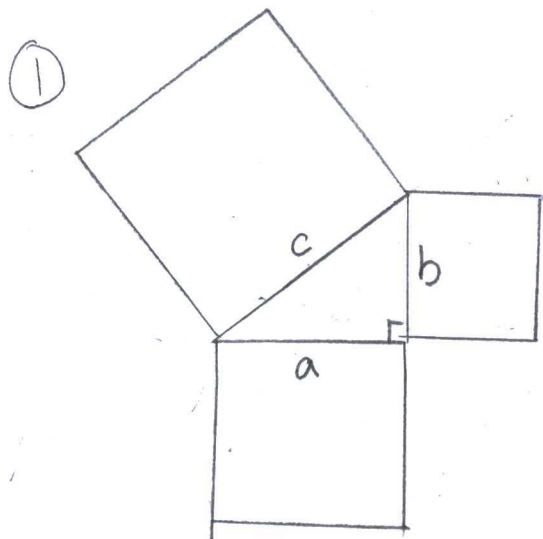


Euclidean and Non-Euclidean Geometry:

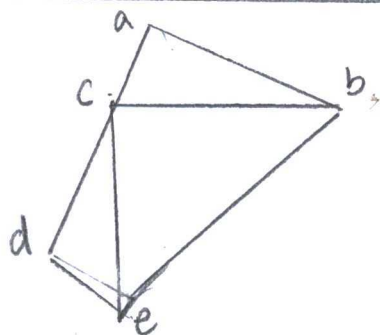
Projects:

- 1) a) Three proofs of the Pythagorean theorem:



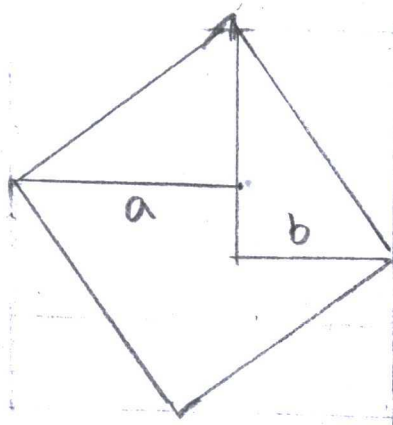
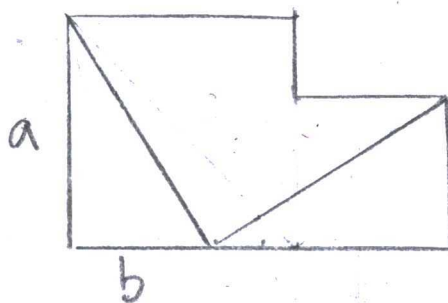
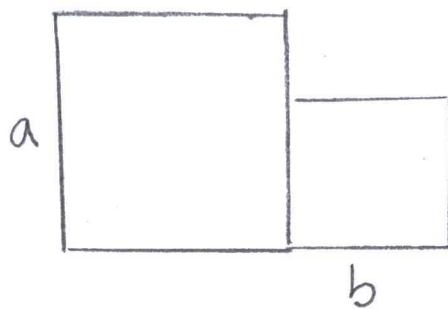
$$a^2 + b^2 = c^2$$

② cite: Garfield, James
(20th President of U.S.A.)
"Correspondence"
New England Journal of
Education. Vol. 13. No. 14 (1876)



$$\begin{aligned} & (a+b) \times \frac{1}{2}(a+b) \\ &= \frac{1}{2}(a \times b) + \frac{1}{2}(c \times c) + \frac{1}{2}(a \times b) \\ & a^2 + b^2 = c^2 \end{aligned}$$

③

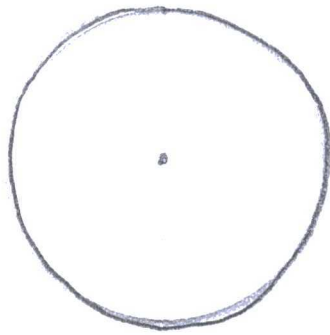


2. Five Euclidean relationships proved:

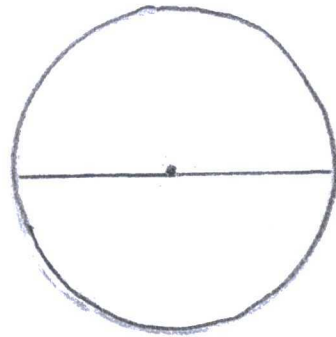
① (pg 31) "III. 31"

"An angle with vertex on a circle and subtending a semi-circle of that right angle."

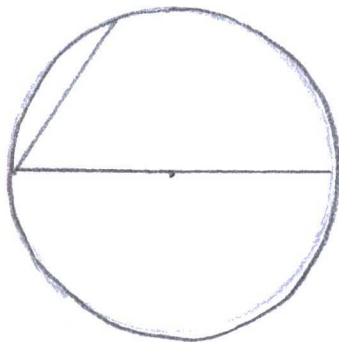
①



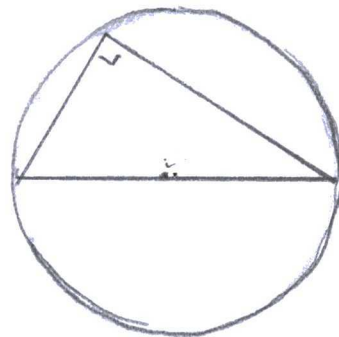
②



③



④



② (pg 31) "I. 34"

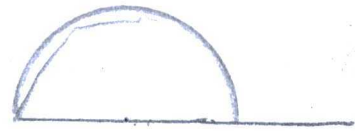
"Opposite sides and angles of a parallelogram are congruent, respectively."

→

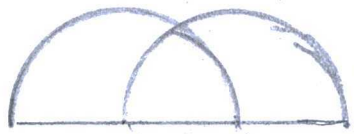
①



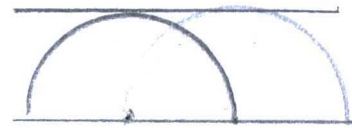
②



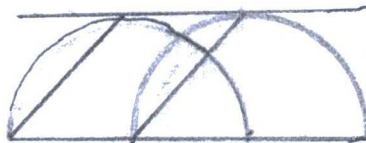
③



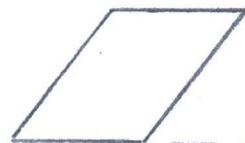
④



⑤



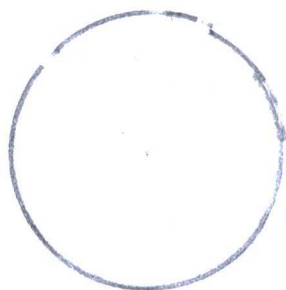
⑥



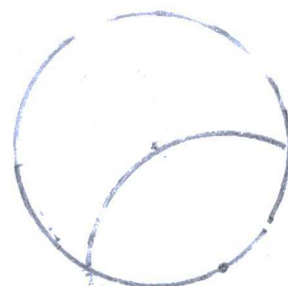
③ (pg 30) "IV.15"

"Inscribe a regular hexagon in a circle"

①



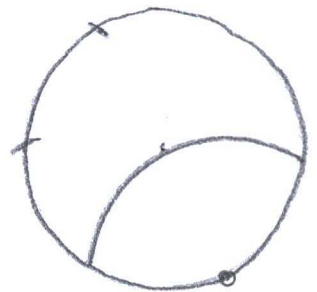
②



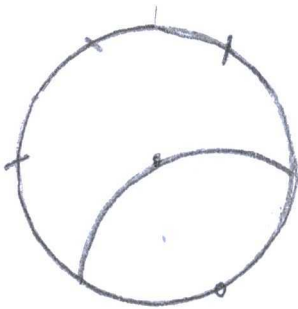
③



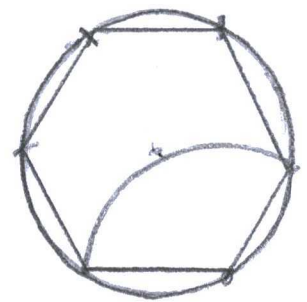
④



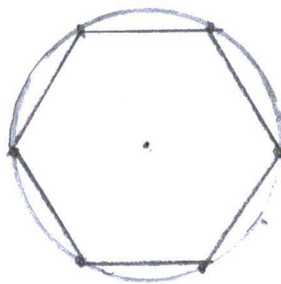
⑤



⑥



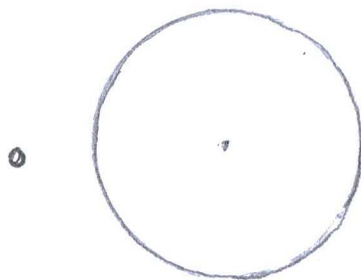
⑦



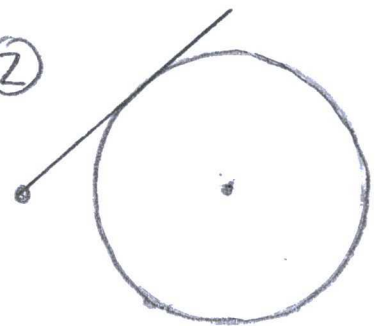
④ (pg 30) "III.17"

"Tangent to a circle from a point outside the circle."

①



②



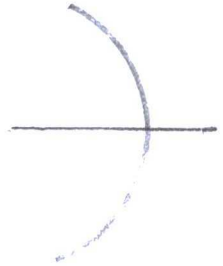
⑤ (pg 30) "I. 1"

"An equilateral triangle in a given segment."

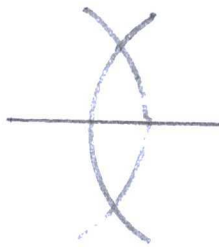
①



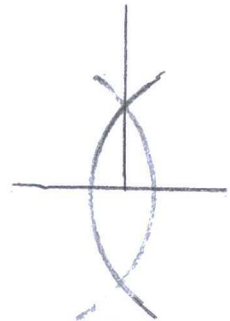
②



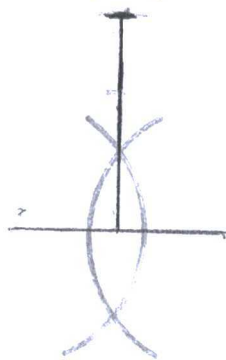
③



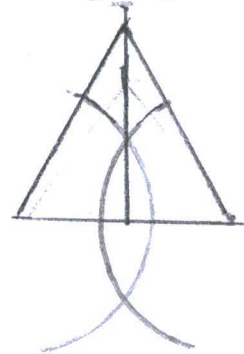
④



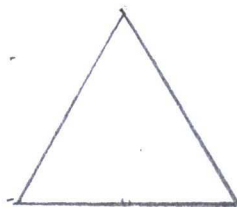
⑤



⑥



⑦



3.

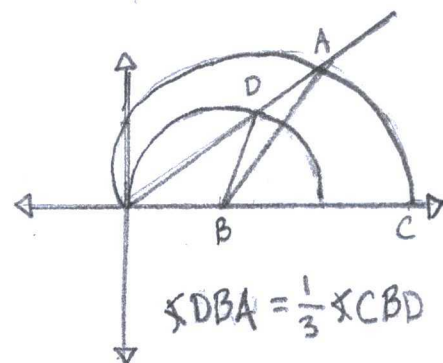
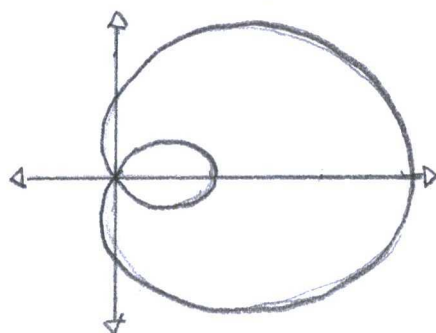
Citation: Moise, E.
 "Elementary Geometry from
 an standpoint"
 University of New York (1990)

Moise spoke about a "surd" angle, such
 as $6.\overline{66}^\circ$. Geometry had problems with
 irrational numbers. $P(\pi)$ was another
 value, also irrationally "surd."

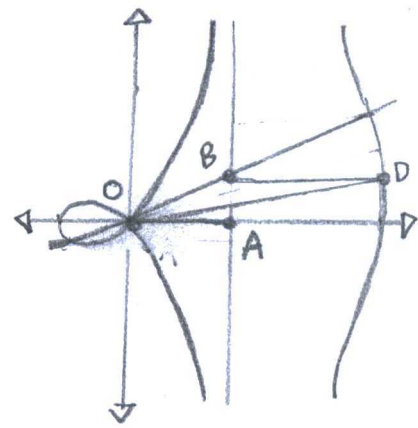
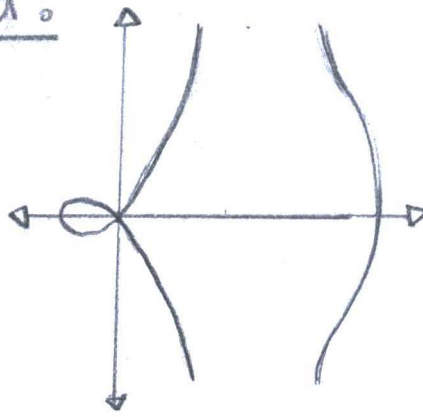
Citation: Peresseni, A., Sherbert D.
 "Topics in Modern Mathematics
 for Teachers."
 University of Illinois (1971)

Angles trisected by:

Limaçon:

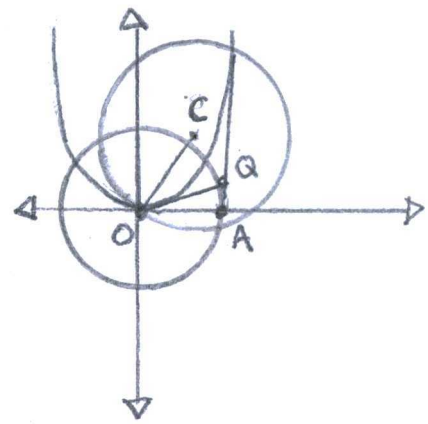
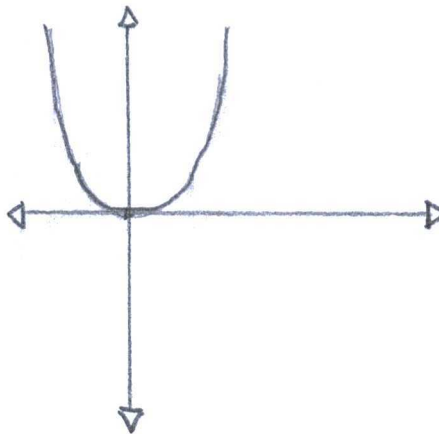


Conchoid:



$$\angle AOD = \frac{1}{3} \angle AOB$$

Parabola:



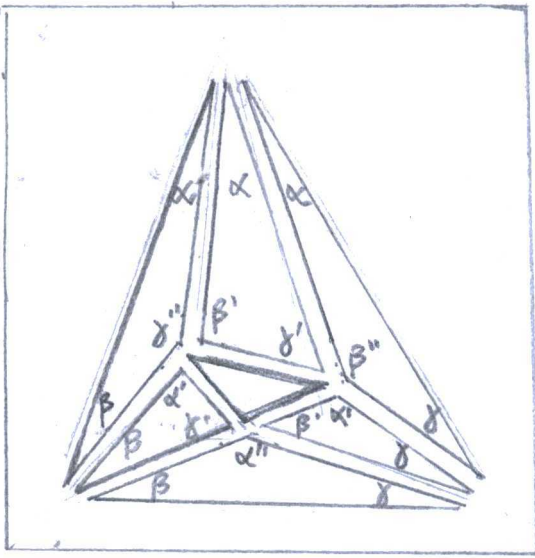
$$\angle AOQ = \frac{1}{3} \angle AOC$$

4

- a) Mathematicians argued about compass versus straight edge by two points on a curve.

- b) Other mathematicians construct with a straight edge alone, only when a single circle and center initially.

5



"Trisecting Angles"

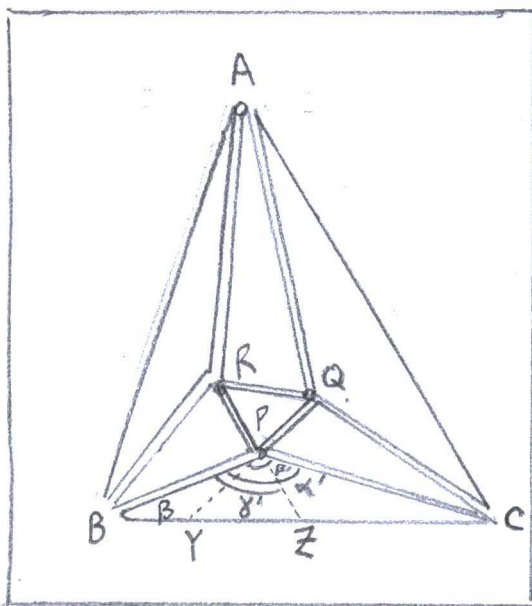
Proof by John Conway (1996)
[Game of Life]:

<u>Hypothesis</u>	<u>Evidence</u>
1) $180^\circ = \beta + \alpha' + \gamma'$ $= \beta + \alpha + \gamma$ $+ 120^\circ$	(pg 31) "I. 17"
2) $180^\circ = \beta' + \alpha' + \gamma$ $= \beta + \alpha + \gamma$ $+ 120^\circ$	(pg 31) "I. 17"
3) $180^\circ = \beta' + \alpha + \gamma'$ $= \beta + \alpha + \gamma$ $+ 120^\circ$	(pg 31) "I. 17"
4) $180^\circ = 3\beta + 3\alpha + 3\gamma$	

Equilateral triangles require equivalent sides and angles, along with rotation.

Angles of Equilateral:

<u>Hypothesis</u>	<u>Evidence</u>
-------------------	-----------------



$$1) \angle PZB = \angle PYC$$

$$2) \angle BPZ = \gamma'$$

$$3) 60^\circ = \beta' + \alpha' + \angle BPZ \quad 60^\circ = \beta' + \alpha' + \gamma'$$

$$4) \angle PYC = \beta'$$

$$5) 60^\circ = \angle PYC + \alpha' + \gamma'$$

$$60^\circ = \beta' + \alpha' + \gamma'$$

$$6) \angle PZB = \alpha'$$

$$7) 60^\circ = \beta' + \angle PZB + \gamma'$$

$$60^\circ = \beta' + \alpha' + \gamma'$$

$$8) \angle PYC = \alpha'$$

$$9) 180^\circ = \beta' + \angle PYC + \gamma'$$

Line #1 from
Constraints above
by 180° degrees
in triangles.

Each angle is 60° ,
as α' , β' , and γ' .

So, equilateral.

Sides in Equilateral:

Hypothesis

Evidence

1) Side $BP = BP$

Euclid Postulate II

2) Angle $\angle ZPB = \angle RPB$

Later chapter 4,
Supplementary angles.

$$3) \angle BZP = \angle BRP$$

Axiom C-4

$$4) \triangle BRP = \triangle BZP$$

Proposition 3.17

(A.S.A. Criterion
for congruence)

$$5) \overline{BP} = \overline{BR}$$

Proposition 3.18

6) Similar argument

Steps #1-5

for $\overline{AR} = \overline{AQ}$ and

$$\overline{CP} = \overline{CQ}$$

$$7) \overline{RP} = \overline{PQ} = \overline{QR}$$

Inner rotation by α'' :

Hypothesis

Evidence

$$1) \overline{BP} = \overline{PC}$$

@ α'' - γ rotation

$$2) \overline{CQ} = \overline{QA}$$

@ β'' - α rotation

$$3) \overline{AR} = \overline{RB}$$

@ γ'' - β rotation

$$4) 360^\circ = \alpha'' - \gamma + \beta'' - \alpha + \gamma'' - \beta$$

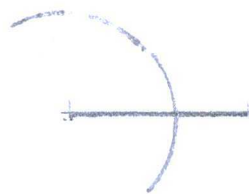
6)

Regular Pentagon by straight edge
and compass:

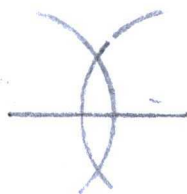
①



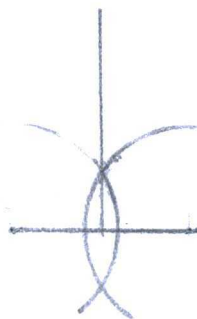
②



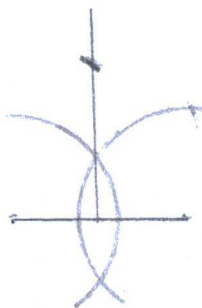
③



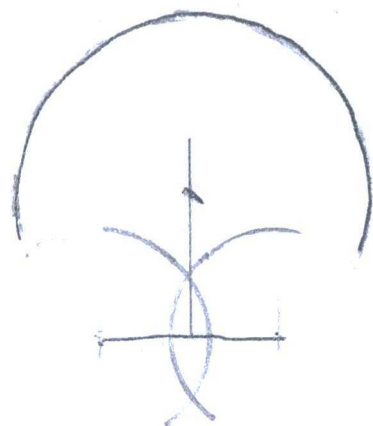
④



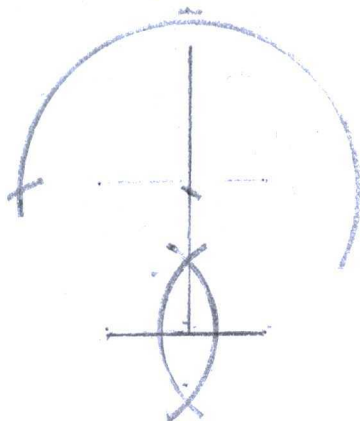
⑤



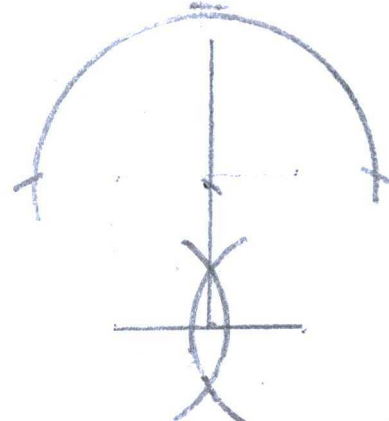
⑥



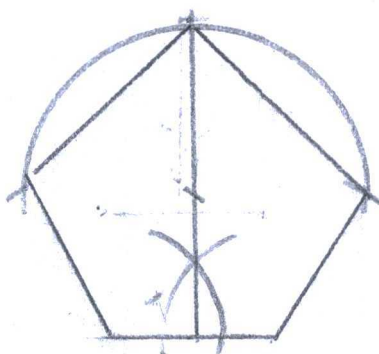
⑦



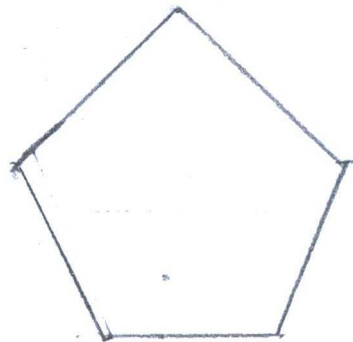
⑧



⑨



⑩



Regular hexagon by straight edge and compass:

①



②



③



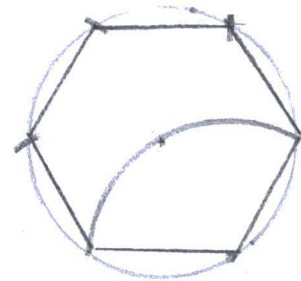
④



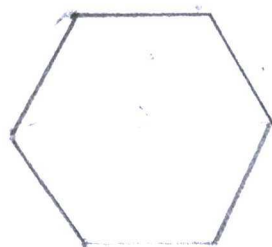
⑤



⑥



⑦



Klein (2007) defined diversity by an image and table. Diversity has types; separation, Variety, and disparity.

7.

Viète was a mathematician, author, magistrate, lawyer, husband, revolutionary, and radical. Supposedly, Viète invented the equal sign ($=$) in algebra, represented variables by letters, and published tables for trigonometry. Common methods in geometry were by compass and straight edge; but Viète marked one distance on the ruler during angle trisection.

8.

Classical Areas:

Cissoid (by parameter):

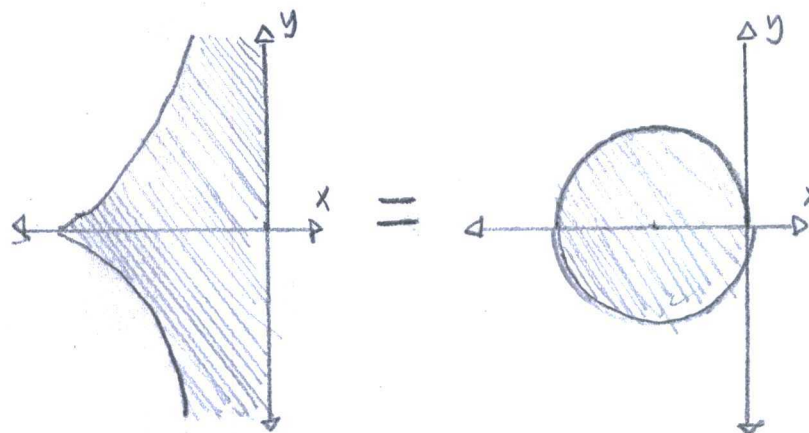
$$\text{Area} = \pi r^2$$

$$= \pi (x^2 + y^2)$$

$$\text{where } x = \frac{2at^2}{1+t^2}$$

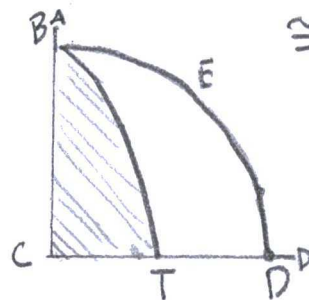
$$y = x \cdot t$$

$a = \text{radius}, t = \tan \theta = \text{slope}$

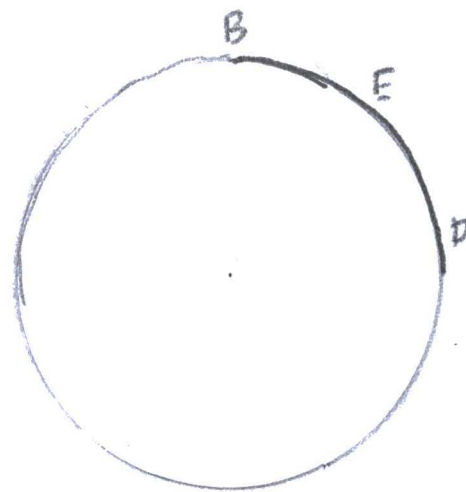


Quadratrix (by curve):

$$\begin{aligned}\text{Area} &= \pi r^2 \\ &= \frac{4(\overline{BED})^2}{\pi} \\ &= \frac{4}{\pi} \left(\frac{\overline{BC}^2}{\overline{CT}} \right)^2\end{aligned}$$



$\cong$

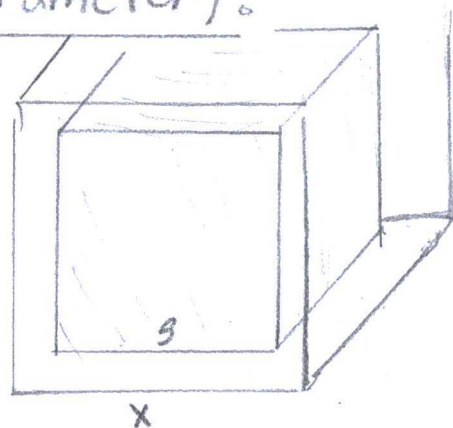


Note: Arc BED on the quadratrix relates to circumference around a circle. Akin to Spirographs, mathematicians built trammels for quadratrix.

Cube Doubling (by parameter):

$$\text{Volume} = s^3$$

$$\begin{aligned}\text{Double volume} &= 2s^3 \\ &= \frac{X}{\sqrt[3]{2}}\end{aligned}$$



9.

Analytic geometry solved problems in existence for 1400 years. From early times, mathematicians; Euler,

Eudoxus had no solution to $x^2=2$.

Numbers with irreducible magnitude ($\sqrt{2}$, $\sqrt[3]{2}$, π , $1/3$) were not possible in geometry by straight edge and compass.

Major problems in geometry never applied series, primes, or algebra. Viète

solved side to a square with area at 2 by a series. Then later,

Descartes solved angle trisection, tangents, and irreducible numbers

from content in analytic geometry.

10.

Eudoxus of Cnidus was from West Asia, now Republic of Türkiye, but in 400 B.C. In Hellenistic

timelines, Eudoxus lived during the

Peloponnesian wars. However, no physical documents exist from Eudoxus.

The poets, critics, and other mathematicians describe Eudoxus, even today Stanford university transliterates Greek from fragments, but with no original tablet or scroll. Popular discoveries by Eudoxus were: days/months/seasons by spheres around Earth, radius proportional to spheres, side lengths on a cube, and ratio.

11. Original rational numbers were not in Base-10, but Base-60. The common number 60 had a multiple of seconds, minutes, a divisible in the Gregorian Calendar, and Chinese calendar. Babylonians described irrational

numbers by successive fractions with 60 in the denominator. Square root two was $1 + \frac{24}{60} + \frac{51}{60^2} + \frac{10}{60^3} + \dots$. Old tablets showed a relationship about irrational numbers not in Base-10.

12. The author asks, "Does explaining a subject in too much detail deprive the reader of pleasure?" I argue, "No." Pleasure derives from family, friends, and neighbors. As a minor, virtues and value motivate pleasure. Too much detail never deprives the reader of pleasure. Family is the source of pleasure in most creatures, not an individual's extensive detail in a topic.

13.

a) (from exercise) "The axiomatic method has many advantages over honest work." - Bertrand Russell

An axiom equals honest work,
but also not a theorem or law.

b) (from exercise) "Our difficulty is not in the proofs, but in learning what to prove." - Emil Artin.

Wrong! Humans know significant problems across computer science, cellular biology, statistics, and astronomy. Millenium Prize Problems are noteworthy. D.A.R.P.A. Mathematical Challenge promise no proof, yet. Also, Erdős (>1123 problems) had no proof.